**UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**GCE Ordinary Level**

# MARK SCHEME for the October/November 2008 question paper

## 5070 CHEMISTRY

**5070/04** Paper 4 (Alternative to Practical), maximum raw mark 60

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the October/November 2008 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.

UNIVERSITY of CAMBRIDGE
International Examinations

**1** **(a)** **(i)** nitrogen [1]

**(ii)** 64 $cm^3$ [1]

**(iii)** 16/80 = 20% [1]

**(b)** **(i)** $2Cu + O_2 = 2CuO$ [1]

**(ii)** black [1]

**(c)** **(i)** 0.16/64 = 0.0025 moles [1]

**(ii)** 0.00125 moles [1]

**(iii)** 30 $cm^3$ [1]

**(iv)** 150 $cm^3$ [1]
(If $dm^3$ used in both **(iii)** and **(iv)** and stated, no deduction.
If not stated mark lost in **(iii)** but e.c.f for **(iv)**.

**[Total: 9]**

**2** **(a)** **(i)** orange to green [1]

**(ii)** oxidising agent etc. [1]

**(iii)** sulfur dioxide or hydrogen sulfide [1]

**(iv)** propanol, [1]

**(b)** **(i)** propyl propanoate (e.c.f on incorrect alcohol in **(a) (iv)**) [1]

**(ii)** esters [1]

**(iii)** sweet or fruity smell (e.c.f for **(i)** and **(ii)** on propene only) [1]

**(c)** yellow or orange (propanoic acid ), red (sulfuric acid) [1]

**(d)** **(i)** gas evolved, effervescence, fizzing, bubbles, Mg dissolves, or test-tube gets hot. [1]

**(ii)** reaction faster with sulfuric acid [1]

**(e)** sulfuric acid is a stronger acid (than propanoic acid) [1]

**[Total: 11]**

© UCLES 2008

**3** (c) [1]

**4** (d) [1]

**5** (b) [1]

**6** (b) [1]

**7** (b) [1]

**[Total 3–7: 5]**

**8** **(a)** 1.32 g [1]

**(b)** **(i)** 106 g [1]

**(ii)** 1.32 × 4 / 106 = 0.0498 (0.05) mol/$dm^3$ [1]

**(c)** **(i)** yellow to **(ii)** orange, red, pink. [1]

**(d)** titre was too small to obtain accuracy etc [1]

**(e)** water – first (1) diluted acid or solution H – second (1) [2]

**(f)**

| | | |
|---|---|---|
| 23.7 | 40.6 | 44.2 |
| 0 | 17.5 | 20.9 |
| 23.7 | 23.1 | 23.3 |

Mean value = 23.2 (1) $cm^3$

(1 mark for each correct row or column (3)) [4]

**(g)** 0.00125 [1]

**(h)** 0.0025 [1]

**(i)** 0.0025 × 1000 / 23.2 = 0.108 [1]

**(j)** 1.08 mol/$dm^3$ [1]

**(k)** Increase the concentration or amount of $Na_2CO_3$ (1)
by factor of 10 (1) [2]

**[Total: 17]**

© UCLES 2008

**9** **(a)** Colourless solution (1)

**(b)** $Al^{3+}$ (1) and $Zn^{2+}$ (1) (or any correct ion e.g. $Pb^{2+}$)
(If either or both charges are incorrect or missing -1)

**(c)** No precipitate or slight white ppt. (1) (<u>not</u> no reaction)

**(d)** $HNO_3$ (not conc) (1) $Pb(NO_3)_2$ or $AgNO_3$ (1)
yellow ppt. (1)
$CaI_2$ (1) [8]

**[Total: 8]**

**10** **(a)** 29, 49, 21, 18. (1) all correct.
37, 33, 28, 29. (1) all correct.
12, 8, 3, 4. (1) all correct. [3]

**(b)** Points connected by a smooth curve [1]

**(c)** **(i)** 10 $cm^3$ (read candidates graph). (Must show evidence of extending graph). [1]

**(ii)** The greater the atomic mass of the element the less moles or amount are/is involved.
(or w.t.t.e) [1]

**(d)** Points connected by a series of straight lines (1)
all points plotted correctly in <u>both</u> graphs (1) [2]

**(e)** graph does not show any relationship between the two,
not uniform, not a curve or a straight line, or w.t.t.e. [1]

**(f)** Copper does not react with hydrochloric acid. [1]

**[Total: 10]**

© UCLES 2008